

Yaghoub Rahimi

Postdoctoral Researcher in Mathematics

Department of Mathematics, Louisiana State University

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Academic Appointments

Louisiana State University

August 2025–Present

Postdoctoral Researcher in Mathematics, Baton Rouge, Louisiana

Research in analytic number theory and discrete harmonic analysis.

Education

Ph.D. in Mathematics

August 2025

Georgia Institute of Technology, Atlanta, Georgia

Dissertation: *Improving averages over the prime numbers and Goldbach's conjecture*. Advisor: Michael T. Lacey.

M.S. in Mathematics

August 2016–July 2019

The University of Texas at Dallas, Richardson, Texas

Advisor: Yifei Lou.

M.S. in Mathematics

September 2013–January 2016

Sharif University of Technology, Tehran, Iran

Advisor: Ali Daneshgar.

B.S. in Applied Mathematics

September 2009–2013

Shiraz University, Shiraz, Iran

Research Interests

Analytic number theory; discrete harmonic analysis; additive combinatorics; prime numbers; Hardy–Littlewood circle method; arithmetic progressions; restricted-digit problems.

Publications

Published and Accepted Articles

1. M. T. Lacey, H. Mousavi, **Y. Rahimi**, and M. N. Vempati, “A density theorem for higher-order sums of prime numbers,” *Research in the Mathematical Sciences* **13** (2026), Article 17. Journal | arXiv
2. **Y. Rahimi**, S. H. Kang, and Y. Lou, “A lifted ℓ_1 framework for sparse recovery,” *Information and Inference: A Journal of the IMA* **13** (2024), no. 1, iaad055. Journal | arXiv
3. M. T. Lacey, H. Mousavi, and **Y. Rahimi**, “Endpoint ℓ^r improving estimates for prime averages,” *Mathematical Research Letters* **29** (2022), no. 6, 1767–1791. Journal | arXiv
4. C. Giannitsi, M. T. Lacey, H. Mousavi, and **Y. Rahimi**, “Improving and maximal inequalities for primes in progressions,” *Banach Journal of Mathematical Analysis* **16** (2022), Article 42. Journal | arXiv
5. C. Wang, M. Yan, **Y. Rahimi**, and Y. Lou, “Accelerated schemes for the L_1/L_2 minimization,” *IEEE Transactions on Signal Processing* **68** (2020), 2660–2669. Journal | arXiv
6. **Y. Rahimi**, C. Wang, H. Dong, and Y. Lou, “A scale-invariant approach for sparse signal recovery,” *SIAM Journal on Scientific Computing* **41** (2019), no. 6, A3649–A3672. Journal | arXiv

Preprints

7. R. Han and **Y. Rahimi**, “On the small denominator problem for generalized Minkowski–Funk transforms,” preprint. arXiv
8. C. Giannitsi, B. Krause, M. T. Lacey, H. Mousavi, and **Y. Rahimi**, “Averages over the Gaussian primes: Goldbach's conjecture and improving estimates,” preprint. arXiv

Selected Research Talks

- A Density Theorem for Higher-Order Sums of Prime Numbers** *March 2025*
Southern Regional Number Theory Conference, Louisiana State University, Baton Rouge, Louisiana.
- Averages over the Gaussian Primes: Improving Estimates and Goldbach's Conjecture** *February 2024*
Number Theory Seminar, Louisiana State University, online.
- Explicit Construction of RIP Matrices and Related Problems** *October 2023*
Heat Flow, Two-Point Inequalities, and Beyond Summer School, online workshop.
- Endpoint ℓ^p Improving Estimates for Prime Averages** *May 2021*
Combinatorial and Additive Number Theory Seminar (CANT), online seminar.
- A Scale-Invariant Approach for Sparse Signal Recovery** *Spring 2019*
Poster presentation, The University of Texas at Dallas, Richardson, Texas.

Teaching Experience

Instructor of Record

- MATH 1554: Linear Algebra**, Georgia Institute of Technology *Fall 2023*
MATH 2551: Multivariable Calculus, Georgia Institute of Technology *Summer 2023*

Teaching Assistant

- Georgia Institute of Technology** *Fall 2019–Spring 2025*
Differential Equations; Abstract Algebra I–II; Multivariable Calculus.
- The University of Texas at Dallas** *Fall 2016–Summer 2019*
Topology; Differential Calculus; Applied Calculus I–II; Calculus I–II.
- Shiraz University** *Fall 2010–Spring 2013*
Foundations of Geometry; Linear Algebra I; Introduction to Computing and Programming; Elementary Matrix and Linear Algebra; Foundations of Mathematics.

Honors and Awards

- Outstanding Teaching Assistant Award, Georgia Institute of Technology, April 2025.
- Research Prize, supported by NSF Grant DMS-1949206, Georgia Institute of Technology, Spring 2021.
- David L. Brown Fellowship Award, Georgia Institute of Technology, April 2020.
- Bronze Medal, 20th International Mathematics Competition for University Students, Blagoevgrad, Bulgaria, August 2013.
- Gold Medal, 37th Iranian Mathematics Competition, Semnan University, Iran, February 2013.
- Fifth Place, 18th Iranian Mathematics Olympiad for University Students, Tarbiat Modares University, Tehran, Iran, July 2013.
- Thirteenth Place, ACM International Collegiate Programming Contest, Asia Regional – Tehran Site, December 2012.
- Second Place, Shiraz ACM-ICPC Contest, Shiraz University, October 2012.

Mentoring and Academic Service

- Mentor, Directed Research Program, The University of Texas at Dallas, Fall 2018, Spring 2019, and Fall 2019.
- Member, Organizing Committee, Workshop on Recent Developments in Mathematical and Statistical Approaches in Data Science, The University of Texas at Dallas, May 31–June 2, 2019.
- Coach, Shiraz University team, 39th Iranian Mathematics Competition, Yazd University, May 2015.
- Mathematics Olympiad Instructor, high schools in Shiraz, Iran, 2012.

Professional Membership and Technical Skills

Professional membership: Member, American Mathematical Society, 2019–Present.

Technical skills: $\text{\LaTeX}$; Python; MATLAB; Mathematica; R; C/C++.